

560. Subarray Sum Equals K Date: _____

Given an array of integers `nums` and an integer `k` return the total number of subarrays whose sum equals to `k`.

* SubArray is a contiguous non-empty sequence of elements within an array.

Example 1:-

Input: `nums = [1, 1, 1]`, `k = 2`

output: 2

Explanation $\text{nums}[0] + \text{nums}[1] = 2$
 $\text{nums}[1] + \text{nums}[2] = 2$ ⓧ ②
Sub Array

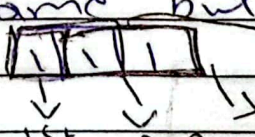
Example 2:-

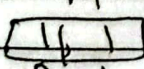
Input: `nums = [1, 2, 3]`, `k = 3`

output: 2

Explanation:- $\text{nums}[0] + \text{nums}[1] = 3$ total
 $\text{nums}[2] = 3$ ②
Sub A

* we can solve it by Brute force method like we can find ~~&~~ All subarrays sum then if it equal to `k` Count ~~to~~ increase by 1 but the time comp is $\sim O(N^3)$ or with better solution $\sim O(N^2)$

* Better solution same but ~~&~~ with two loop like `nums`  1st 2nd 3rd subarrays

* we can find their sum by two loop like suppose  $\rightarrow$ 1st subarray sum = 1
 $\rightarrow$ 2nd subarray sum = 2

"we have to Add '1' index value = 1 in sum 0 index value" are present in sum.

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* Optimal Solution:-

nums = [1, 2, 3, -3, 1, 1, 1, 4, 2, -3] K = 3

* If we find sum of limited Range
 $\Rightarrow$ Suppose on 6 Index

[1, 2, 3, -3, 1, 1, 1] ~~2, 4, 2, -3~~

$$1 + 2 + 3 + (-3) + 1 + 1 + 1 = 6$$

[1, 2, 3, -3] [1, 1, 1] $\rightarrow$ total sum = 6

PrefixSum = nums[0] + nums[1] + nums[2] + nums[3] = total sum - K
 right

* If we count [total sum - K] mean we can find ^{sum of} SubArray = K

* Why count 2 may be [total sum - K] are more than 1 these are prefix sum

[1, 2, 3, -3, 1, 1, 1] $\rightarrow$ total sum = 6

PrefixSum = nums[0] + nums[1] + nums[2] + nums[3] + nums[4] + nums[5] + nums[6] = total sum - K

* see total sum = 6 with same number of Index but number prefix (total sum - K) = 2

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* which data structure we will use and why?

⇒ we will use Hashmap b/c we need to count of prefixSum and prefixSum may duplicate so we will set (Keys, value) → Keys = prefixSum
Values = Count.

like in back side ~~expt~~ example prefixSum=3
→ their count = 2 mean duplicate prefixSum
just we have to increase the count
b/c Keys can not be duplicate.

⇒ If we found Keys have ~~2~~ values
~~or 2 values~~ mean there are ~~2~~
number of SubArray.

Dry Run Example $nums = [1, 2, 3], K = 3$

steps	elements	currsum	currsum-k	HashMap	Count
init	—	—	—	{0:1}	0
1	1	1	-2	{0:1, 1:1}	0
2	2	3	0	{0:1, 1:1, 3:1}	1
3	3	6	3	{0:1, 1:1, 3:1, 6:1}	2

→ this is b/c ^{be} may input is one
number like {3}

$K = 3$

Current Sum = 3

Current Sum - K = 0

→ already
in Hash
map.